CS3081 · ARTIFICIAL INTELLIGENCE

RESOLUTION HACKATHON

> case_file: the_missing_exam_paper

In-class team sprint · 50 min · 4 levels · 100 XP + bonus

Mentor: Dr. Naila Marir

incident_report.txt

```
$ cat incident_report.txt

Monday 08:00. The CS3081 final exam paper is gone from the instructor's office.
Three students were seen near the office on Sunday night:
  → Sara   → Omar   → Lina

Campus security left 7 clues. Your team has 50 minutes to hack the case with
resolution:
find who took it, who is innocent, and which accusation cannot be proven.

$ ./start_hackathon --teams 3-4 --mode proof_by_contradiction
```

// TEAM REGISTRATION

TEAM NAME		TABLE #	
HACKER 1	Noor Alkibsy	HACKER 2	Tala Ahmed Alzaeem
HACKER 3	Zainab Alamari	HACKER 4	

// THE 4 LEVELS

LEVEL 01
CNF Relay
10 min · 20 XP

LEVEL 02
Resolution Dominoes
10 min · 30 XP

LEVEL 03
Solve the Case
20 min · 40 XP

LEVEL 04
Final Accusation
10 min · 10 XP

Bonus XP: +3 FIRST BLOOD per level · +3 BUG HUNTER per accepted objection · +3 SIDE QUEST in Level 03

README.md

Rules of the Hack

50 MIN | 100 XP

▶ Pick your roles — rotate every level

C
CONVERTER
Turns logic into CNF: eliminate ↔ and →, De Morgan, distribution.

R
RESOLVER
Hunts complementary literals and builds new clauses (resolvents).

✓
CHECKER
Debugs every step and writes the final answer. Nothing ships without the Checker.

▶ How the game runs

01 Teams of 3–4. Play the four levels in order.

02 When your team clears a level, raise your hand. The mentor checks it and unlocks the next level.

03 **First Blood:** the first team to clear each level correctly gets **+3 XP**.

04 Allowed moves only: eliminate ↔, eliminate →, De Morgan, distribution, resolution and factoring.

05 Show every step. A correct answer with no steps earns only half the XP.

▶ XP table

LEVEL	CHALLENGE	TIME	XP
01	CNF Relay — convert the clues	10 min	20
02	Resolution Dominoes — match clause cards	10 min	30
03	Solve the Case — proof by contradiction	20 min	40
04	Final Accusation — present and object	10 min	10 + steals
	Total	**50 min**	**100 + bonus**

DATASET

The Evidence Board

10 SYMBOLS | 7 CLUES

▶ Symbol table

SYMBOL	MEANING
D	The office door was unlocked
S	Sara took the exam paper
O	Omar took the exam paper
L	Lina took the exam paper
C	The security camera was switched off
F	The office door was forced open
Y	Lina had a copy of the office key
K	Khalid lent his office key
A	The alarm rang

▶ Clues from campus security

#	CLUE (ENGLISH)	LOGIC
1	If the door was unlocked, then Sara or Omar took the paper.	D → (S ∨ O)
2	The door was unlocked.	D
3	If Sara took the paper or the door was forced, then the camera was off.	(S ∨ F) → C
4	The camera was NOT off (it recorded all night).	¬C
5	Lina could only have taken it if she had a copy of the key.	L → Y
6	Lina had a copy of the key if and only if Khalid lent his key.	Y ↔ K
7	It is not true that Khalid lent his key AND the alarm rang.	¬(K ∧ A)

CHEAT CODE

To show KB ⊨ α: add **¬α** to the KB, convert everything to CNF, and keep resolving. If you produce the empty clause **( )**, it's a contradiction, so KB ⊨ α. If no new clauses can be made and there is no ( ), there is **no entailment**.

LEVEL 01 / 04

CNF Relay

10 MIN | 20 XP

RELAY Pass the sheet around the team. **Each hacker writes ONE step, then passes it on.** Write the rule you used next to each step.

▶ Clue 1: D → (S ∨ O) [4 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	¬D∨(S∨O)	implication elimination	Noor Alkibsy
2	¬D∨S∨O	removing parathesis	Noor Alkibsy
3			

▶ Clue 3: (S ∨ F) → C [6 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	¬(S∨F)∨C	Elimination implication	Tala Alzaeem
2	(¬S∨¬F)∨C	De Morgan's law	Tala Alzaeem
3	(¬S∨C)∧(¬F∨C)	Destributive law	Tala Alzaeem

▶ Clue 6: Y ↔ K [6 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	(Y→K)∧(K→Y)	Biconditional elimination	Noor Alkibsy
2	(¬Y∨K)∧(¬K∨Y)	implication elimination	Noor Alkibsy
3	(¬Y∨K)∧(Y∨¬K)	commutative law	Noor Alkibsy

▶ Clue 7: ¬(K ∧ A) [4 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	⌐K∨⌐A	De morgan's law	Zainab Alaweri
2			
3			

▶ Compile the final KB in CNF

List every clause from all seven clues, one per row, as K1, K2, K3, … You need these numbers in Level 03.

NO.	CLAUSE	NO.	CLAUSE
K1	⌐D∨C∨O	K6	⌐L∨Y
K2	D	K7	⌐Y∨K
K3	⌐S∨C	K8	⌐K∨Y
K4	⌐F∨C	K9	⌐K∨⌐A
K5	⌐C	K10	

LEVEL 02 / 04

Resolution Dominoes

10 MIN 30 XP

Cut out the 12 clause cards on the next page and shuffle them. Then:

01 Find pairs of cards that share a **complementary literal** (like P and ¬P). Each card belongs to exactly one pair.

02 Place the two cards side by side like dominoes and write the resolvent below.

03 If the resolvent has a duplicate literal, apply **factoring** and say so.

⚠ BUG One pair is a **trap**. It looks like it gives ( ) but it doesn't. Think carefully about what one resolution step is allowed to do.

PAIR	CARD + CARD	RESOLVED ON	RESOLVENT	FACTORING?
1	card E + Card L (M) (¬H)	M	( ) Empty clause.	Y / (N)
2	card B + card G (¬T ∨ U) (T ∨ U)	T	U ∨ U → U	(Y) / N
3	card C + card H (¬Q ∨ R) (P ∨ Q)	Q	P ∨ R	Y / (N)
4	card D + card J (X ∨ ¬Y ∨ Z) (Y ∨ W)	Y	X ∨ Z ∨ W	Y / (N)
5	card F + card K (A ∨ B ∨ S) (¬A ∨ R ∨ S)	A	B ∨ R ∨ S	(Y) / N
6	card A + card I (G ∨ H) (¬G ∨ ¬H)	G or H	H ∨ ¬H G ∨ ¬G	Y / (N)

▶ Bug report: the trap pair

Which pair is it? Explain in one or two sentences why it does NOT give the empty clause.

pair 6 (Card A + Card I) ∵ Standard resolution allows us to cancel out only one pair of complementary literals per step. resolve the G leaves (H ∨ ¬H) and resolve the H leaves (G ∨ ¬G)

LEVEL 02 · ASSETS

Clause Cards

CUT ON DASHED LINES

CARD_A (G ∨ H) { resolve me }	CARD_B (¬T ∨ U) { resolve me }	CARD_C (¬Q ∨ R) { resolve me }
CARD_D (X ∨ ¬Y ∨ Z) { resolve me }	CARD_E (M) { resolve me }	CARD_F (A ∨ B ∨ S) { resolve me }
CARD_G (T ∨ U) { resolve me }	CARD_H (P ∨ Q) { resolve me }	CARD_I (¬G ∨ ¬H) { resolve me }
CARD_J (Y ∨ W) { resolve me }	CARD_K (¬A ∨ R ∨ S) { resolve me }	CARD_L (¬M) { resolve me }

LEVEL 03 / 04

Solve the Case

20 MIN 40 XP

Use your CNF KB (K1, K2, ...) from Level 01. For each case, add the **negation** of what you want to prove, then resolve. Number new clauses R1, R2, R3, ...

▶ Case 3A — Did Omar take the paper? Prove KB ⊨ O [15 XP]

ADD TO KB:

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K1 (¬D ∨ S ∨ O)	K2 (D)	D	S ∨ O
R2	R1 (S ∨ O)	add (¬O)	O	S
R3	K3 (¬S ∨ C)	K5 (¬C)	C	¬S
R4	R2 (S)	R3 (¬S)	S	( ) Empty
R5	—	—	—	—

VERDICT: Because the Empty clause () was produced, a contradiction is proven therefor KB ⊨ O is True, Omar took the Exam paper.

▶ Case 3B — Is Sara innocent? Prove KB ⊨ ¬S [10 XP]

ADD TO KB:

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	added clause (S)	K3 (¬S ∨ C)	S	C
R2	R1 (C)	K5 (¬C)	C	( ) Empty clause
R3	—	—	—	—

VERDICT: Because the Empty clause () was produced, a contradiction is proven. Therefor. KB ⊨ ¬S is True sara is completely innocent.

LEVEL 03 / 04 · BOSS FIGHT

Case 3C — What about Lina?

15 XP

interrogation.log

```
Omar: "Lina did it with me!"
Lina: "I am innocent!"
$ try prove(L) && try prove(¬L)
```

▶ Attempt 1 — add ¬L and resolve

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	KG(¬L ∨ Y)	added(¬L)	None	(No complimentary literal)
R2				
R3				
R4				

▶ Attempt 2 — add L and resolve

(No resolution can be made with (¬L) No Empty Clause ( ).

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	KG(¬L ∨ Y)	added(L)	L	~~Y~~
R2	R1 (Y)	K7 (¬Y ∨ K)	Y	K
R3	R2 (K)	K9 (¬K ∨ ¬A)	K	¬A
R4				

▶ Conclusion

resolution stop at ¬A No Further clause can be resolved and no Empty Clause ( ).

Can the detectives accuse Lina? Can they clear her? What does this mean about entailment?
Hint: look at where the literal L appears in the KB.

As hinted above, the literal L only appears once in the entire KB (inside KG: ¬L ∨ Y) as negative literal. Since it's unconstrained by any other facts, the truth value of L is completely independent; therefore its impossible to prove her innocence or guilt.

SIDE QUEST Finished early? Prove that the door was **not** forced: KB ⊨ ¬F. Write your proof on the back of this page for **+3 XP**.

step	Clause 1	Clause 2	resolved	New clauses
R1	added (F)	K4 (¬F ∨ C)	F	C
R2	R1 (C)	K5 (¬C)	C	( ) Empty

LEVEL 04 / 04

Final Accusation

10 MIN 10 XP

01 The mentor picks one case (3A, 3B or 3C) for your team. Present it on the board in **2 minutes**.

02 While a team presents, the other teams are **code reviewers**. They look for an invalid step.

03 To object, hold up your **OBJECTION!** card and name the exact step and the broken rule.

04 A **correct objection steals 3 XP** from the presenting team. A wrong objection costs 2 XP.

05 A clean presentation with no accepted objection earns the full 10 XP.

▶ Code review log

AGAINST TEAM	STEP	WHAT WAS WRONG	ACCEPTED?
Team 2	R2	wrong resolution rule	Yes
Team 3	R3	Removed two literals in one step	Yes
Team 4	R1	Incorrect resolvent	No

▶ Post-mortem (individual, one sentence each)

Why is resolution called a “proof by contradiction”? What does it mean when no empty clause can ever be produced?

Q1: Because we added the negation of the goal and try to get the empty clause.

Q2: It means the goal cannot be proven from the KB.

UNLOCKABLES

Achievements & Leaderboard

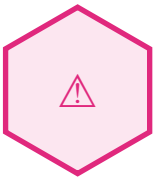
TICK WHAT YOU EARN



FIRST BLOOD

First team to clear a level. +3 XP each time.

□ earned



TRAP DETECTOR

Spotted and explained the trap pair in Level 02.

□ earned



SIDE QUEST

Proved KB ⊨ ¬F in Level 03. +3 XP.

□ earned



CLEAN BUILD

Presented in Level 04 with zero accepted objections.

□ earned

▶ Team scoreboard (mentor use)

L01	L02	L03	L04	BONUS	STEALS	TOTAL
/20	/30	/40	/10	+	±	**/100**

// cut out and hold up during Level 04

ERROR 403: INVALID STEP

OBJECTION!

step ____ breaks rule: _____